INTRODUCTION TO CLOUD COMPUTING
2025 Semester 2

ASSIGNMENT (15%)

The objective of this assignment is to provide students with experience in researching and utilizing a public cloud by installing the web application Wakapi (https://github.com/muety/wakapi). This open-source software is used to collect and summarize time spent coding in various programming languages during project development. Students must install the web application according to the following specifications:

Web Application Installation Requirements on Public Cloud

- Compute Service: The web application must be installed only on a Google Compute Engine Virtual Machine (VM). Students may sign up for Google Cloud to use an e2-micro VM for approximately 744 hours per month (for 31-day months) at no cost. (Please study Google's terms of use at https://cloud.google.com/free/docs/free-cloud-features)
- Restrictions: The use of Google App Engine or Kubernetes is not allowed.
- Database: There are two options:
  1. Use the PostgreSQL Platform-as-a-Service (PaaS) provided by https://neon.com/, which offers 0.5 GB of storage per project for free. (Please study the provider's terms of use).
  2. Use the Google Cloud SQL on Google Cloud Platform. Note that you must be responsible for database security yourself.
- Containers: Installing and using Docker containers on the VM instance is permitted.
- Unmanaged Instance: You must have one unmanaged VM instance (a virtual machine that is not part of a managed instance group).

    - Connectivity: The unmanaged instance must have an external IPv4 address and must be reachable via ping from the Internet.
- Managed Instance Group (MIG): You must create one Managed Instance Group.
    - Instance Template: You must create an Instance Template for the MIG. The web application installation must be performed using a **startup-script-url** only, and this URL must reference a script file stored on Google Cloud Storage.
    - MIG Availability: The Managed Instance Group must have at least one active VM instance at all times.
    - Autoscaling: Support autoscaling based on CPU utilization with a target utilization of 95% (0.95).
    - Scaling Limits: The autoscaling configuration must support increasing the count to a maximum of 2 VM instances.
- Networking: The unmanaged instance and the Managed Instance Group (MIG) must be connected to different Virtual Private Clouds (VPCs). However, all VM instances must be able to ping each other's internal IPv4 addresses.
- Load Balancing: Implement load balancing to distribute traffic between the unmanaged instance and the instances within the Managed Instance Group.
- Monitoring: In Cloud Monitoring, create a custom dashboard to display the average CPU load calculated over a 1-minute interval.
- Alerting: Set an alerting policy so that when the average CPU load exceeds 95%, a notification is sent to parinya.ek@kmitl.ac.th. The email message must include the student’s ID and the specific message provided by the instructor at the time of submission.
- Testing: Test the web application by logging in with admin privileges and successfully changing the Time Zone.

Submission Requirements

- Group Size: This assignment can be done individually or in groups of two. If working in a pair, both students must be present during the on-site submission.
- Deadline: Submissions must be made before March 15, 2026.
- Evaluation: The instructor may require you to create a new user or stop certain VM instances to test the system's functionality.
- Interview: Students must be able to answer questions regarding the installation or configuration. Failure to answer reasonably will result in no points for that specific section.